

Chapter 86 — Urethral Stricture Disease

GENERALITIES OF RECONSTRUCTIVE SURGICAL TECHNIQUES

- **Governing aim: minimise tissue injury and promote healing.** Unlike extirpative surgery, reconstructive results depend on methods that minimise tissue damage and maximise wound healing.
- **Visualisation: loupe magnification** is used by almost all surgeons in adult and paediatric genital reconstruction; a **headlight or lighted suction** helps in deep perineal surgery.

MONOPOLAR VERSUS BIPOLAR CAUTERY

	Monopolar	Bipolar
Grounding	To a pad	To the opposite tine of the forceps
Field effect	Wider	More confined
Use	Superficial structures	Haemostasis around the corpus spongiosum; elevation of penile and scrotal flaps; division of the perineal intracorporal space; dissection of dorsal neurovascular structures; fossa navicularis reconstruction; penile curvature correction; oral mucosal graft harvest

- **Why bipolar near delicate structures:** current is dissipated along conductors — in tissue, the vessels and nerves — so spreading current can injure exactly the structures reconstruction depends on.
- **Instruments:** plastic surgery or peripheral vascular tray — fine tenotomy scissors, fine forceps, skin hooks, delicate needle holders. **Sharp scissors** that cut with minimal collateral trauma are essential.
- **Calibration:** a set of **bougie à boule** sizers checks luminal calibre; McCrea sounds supplement van Buren sounds, but **sounds do not replace bougie à boule calibration.**
- **Posterior urethral reconstruction:** a sound passed through the cystostomy tract and prostate locates the proximal urethral end — **Haygrove sound** (authors), or cystoscope or van Buren sounds.
- **Suture: absorbable is the rule** in urethral surgery — braided absorbable or monofilament absorbable; **chromic rarely** (others superior). Use the **smallest calibre that aligns tissue tension-free**; there is no reason for suture stronger than the tissue.

SUTURE AND NEEDLE SELECTION

Task	Suture	Needle
Flap or graft repair	4-0 to 6-0	Taper where possible (RB-1, TF, SH-1)
Primary anastomosis of spongiosum and urethra; posterior urethral reconstruction	3-0 to 4-0	Taper; UR-6 half-circle taper (as in radical prostatectomy) for deep perineal anastomosis
Severe spongiofibrosis or scar	—	Cutting (P-3, PC-3) when tissue quality prohibits taper

- **Position:** supine or prone where possible; many former lithotomy procedures are done in **frog-leg or split-leg**. **Lithotomy or exaggerated lithotomy only for the minimum time necessary.** With foot padding and no pressure on the back of the leg, low-lithotomy complications are minimal; venous compression stockings in supine, split-leg and low-lithotomy positions.
- **The controversy is exaggerated lithotomy:** the authors use it for **all bulbar and posterior reconstructions** — safe, with unequalled access to deep perineal structures; others use lower lithotomy.

- **Retraction:** Scott retractor with stay hooks (penile surgery); Jordan–Bookwalter perineal retractor set; Omni-Tract perineal retractor.
- **Key ingredients:** visualisation, suture choice, delicate handling, positioning, retraction — on a foundation of **proper diagnosis and planning**.

KEY POINTS: RECONSTRUCTIVE SURGICAL TECHNIQUES

- All efforts aim at **minimising tissue injury and promoting healing**; loupes almost universal; headlight or lighted suction for deep exposure; delicate instruments because sutures and needles are small.
- In urethral surgery **absorbable suture is the rule**, smallest calibre that aligns tissue tension-free.
- **Positioning is the surgeon's preference**. Proper diagnosis and planning are central to success.

Concept — *reconstructive outcome is a wound-healing outcome: magnification, confined current, the finest absorbable suture and a taper needle all serve to limit collateral injury, because the scar the surgeon creates is the next stricture.*

PRINCIPLES OF RECONSTRUCTIVE SURGERY

- Covered in the online version of the chapter and in **Chapter 85**.

SURGICAL ANATOMY OF THE PENIS AND MALE PERINEUM

- Online (Figs 85.5–85.13); full anatomic description in **Chapters 66 and 85**. Key points for stricture surgery are re-emphasised under Urethral Anatomy below.

SELECTED PROCESSES

- **Cross-referred to Chapter 85:** urethral haemangioma; reactive arthritis (Reiter syndrome); lichen sclerosus; amyloidosis; urethrocutaneous fistula; male urethral diverticulum; paraphimosis, balanitis and phimosis. **Circumcision:** Chapters 85 and 55. **Transitional urological care for failed hypospadias and epispadias repair:** online.

Urethral Meatal Stenosis

MEATAL STENOSIS: CHILD VERSUS ADULT

	Child	Adult
Cause	Consequence of circumcision → ammoniacal meatitis (wet diaper pressing on the glans tip) → ventral meatal fusion	Inflammation; specific or non-specific urethritis; trauma (indwelling catheter, instrumentation, some radical prostatectomies); failed hypospadias repair
Lesion	Thin occluding skin membrane, ventral	Rarely isolated — fossa navicularis usually involved
Treatment	Ventral meatotomy (general anaesthesia preferred) + parental dilation	Ventral meatotomy with mucosal sutures; dorsal meatotomy (Y-V-plasty) if needed; evaluate as stricture disease

Child

- A small neonatal meatus rarely reaches a urologist unless it accompanies **other anomalies (hypospadias)** or causes **voiding difficulty or urinary tract infection**.

- **Meatotomy if narrow and symptomatic.** Observe voiding: the meatus should open with a full forceful stream; a **narrow, excessively forceful stream** indicates stenosis; the thin membrane may pouch out, the meatus opening at the dorsal lip.
- **Ammoniacal meatitis:** meatal dilation with **0.05% clobetasol cream** — abates within a week; anecdotally the ventral fusion can be avoided. Counsel parents on the cause.
- **Ventral meatotomy:** local anaesthesia sometimes possible; **general anaesthesia preferred in the young child.** Edges re-seal unless kept open → parents separate them with the **tip of a meatal dilator three times daily for 7–10 days** with copious ointment; the surgeon watches the parents do it. An ophthalmic antibiotic tube tip works, its ointment serving as lubricant.

Adult

- **Ventral meatotomy** in adolescents and adults usually needs **three sutures** (apex and each side) approximating mucosa to skin for haemostasis; a meatal dilator (adult 6–34 Fr, paediatric 6–10 Fr) keeps it open.
- **Dorsal meatotomy** is sometimes needed — a **Y-V-plasty** after excising any scarred ridge of neourethra; effective but **cosmetically suboptimal** meatus.

Concept — *in the child, meatal stenosis is a thin post-circumcision membrane cured by meatotomy and dilation; in the adult it is usually the visible tip of a fossa navicularis stricture and is evaluated as stricture disease.*

KEY POINTS: SELECTED PROCESSES

- Paediatric meatal stenosis follows circumcision → ammoniacal meatitis → ventral membrane. **Neonatal circumcision remains controversial;** if performed, it must be performed properly.
- **Failed hypospadias repair** is complex; many patients are victims of the technology of their era. **Any urethral involvement is evaluated as urethral stricture disease.**
- Modern **exstrophy-epispadias** reconstruction gives better function and less secondary surgery; secondary reconstruction targets the escutcheon, dorsal penile base, shaft, urethra and penoscrotal junction.

URETHRAL STRICTURE DISEASE

- **Urethral stricture = fixed anatomic narrowing** such that the lumen will not accept instrumentation **without disrupting the mucosa.**
- **Anterior stricture:** within the urethra surrounded by corpus spongiosum; scarring may extend through the spongy erectile tissue (**spongiofibrosis**) and into adjacent tissue. **Contraction of this scar reduces the lumen.**
- **Area falls with the square of diameter:** 30 Fr = 10 mm diameter $\approx 78 \text{ mm}^2$; 15 Fr = 5 mm $\approx 20 \text{ mm}^2$. **Halving the calibre removes about three-quarters of the lumen** — so scar contraction can be silent for a while, then produce marked symptoms as the lumen narrows further.

TERMINOLOGY (WORLD HEALTH ORGANIZATION 2004; INTERNATIONAL CONSULTATION ON UROLOGICAL DISEASES, MARRAKESH 2010)

Site	Correct term	Nature
Anterior (spongiosum-invested) urethra	Stricture — the term is reserved for this segment	Epithelial scar $\pm$ spongiofibrosis

Site	Correct term	Nature
Membranous urethra after pelvic fracture	Distraction defect ; posterior urethral injury (PUI)	Obliterative fibrosis between distracted ends
Posterior urethra after trauma or radical prostatectomy	Posterior urethral stenosis	Obliterative fibrosis; defect may be long, tissue process usually confined
Other posterior narrowing (prostatic urethra, bladder neck)	Contracture or stenosis	—

Figure 86.1 — Anatomy of anterior urethral strictures (from Jordan, 1987)

- Most anterior strictures include underlying spongiofibrosis. Spectrum: **(A) mucosal fold → (B) iris constriction → (C) full-thickness, minimal spongy fibrosis → (D) full-thickness spongiofibrosis → (E) inflammation and fibrosis beyond the corpus spongiosum → (F) complex stricture with fistula**, which may progress to abscess or open to skin or rectum.

Urethral Anatomy

- **Bulbar urethra is eccentric** within the corpus spongiosum, much **closer to the dorsum**; distally the **penile urethra becomes central**.
- **Genital skin**: dual (proximal and distal) and bilateral supply → a **fasciocutaneous system**.
- **Corpus spongiosum**: from the **common penile artery**, terminal branch of the internal pudendal artery. **Dual supply**: proximal (arteries to the bulb) and **retrograde through the dorsal arteries arborising in the glans**.

Concept — the **retrograde glanular supply** is what allows the bulb to be transected and mobilised; where it is lost (*hypospadias*, *common penile artery injury*), mobilisation becomes ischaemia — the thread that runs through vessel-sparing excision and pelvic fracture repair failure.

Aetiology

- Any process that injures the epithelium or underlying spongiosum enough to heal with scar causes stricture. Categories: **iatrogenic, traumatic, inflammatory, idiopathic**.
- **Meta-analysis: idiopathic and iatrogenic are the commonest** (roughly equal), followed by post-traumatic, then inflammatory.

AETIOLOGY OF URETHRAL STRICTURE

Category	Mechanisms and examples	Points
Iatrogenic	Diagnostic cystoscopy, urethral dilation; transurethral resection of bladder tumour (TURBT) and of the prostate (TURP) — direct mucosal insult or inflammation → ischaemia (technique, instrument calibre, duration); perioperative catheter — material, insertion technique, dwell time → injury, pressure necrosis, inflammation or infection	Catheter strictures very prevalent in low- and middle-income countries (LMICs); silicone catheters and hydrophilic-coated intermittent catheters may reduce them
Traumatic	Blunt or penetrating; straddle and deceleration — spongiosum crushed against the inferior pubic rami (bulbar); pelvic fracture — obliterative posterior defect; prostate cancer treatment — posterior stenosis	Straddle injury often unrecognised until obstructive symptoms; most straddle bulbar injuries are reconstructible
Inflammatory	Gonorrhoea (historically commonest, now uncommon with prompt antibiotics); lichen sclerosus (LS)	<i>Chlamydia</i> and <i>Ureaplasma urealyticum</i> (non-specific urethritis): no established association

Category	Mechanisms and examples	Points
Idiopathic	No identified cause	Fewer iatrogenic strictures now: small endoscopes, limited cystoscopy indications in young men
Congenital	Stricture at an embryological fusion point (posterior–anterior junction)	Rarest (see below)

- **Lichen sclerosus:** definite association with inflammatory stricture. Begins as **glans inflammation** and **inevitably causes meatal stenosis**, if not a fossa navicularis stricture; cause unknown.
- **Proposed progression in LS:** high-pressure voiding → **intravasation of urine into the glands of Littre** → glandular inflammation, microabscesses → deep spongiofibrosis. Bacterial role undefined; **antibiotics may limit obstructive symptoms but do not resolve the stricture**.
- **Congenital stricture — authors' criteria:** not inflammatory, short, no history of or potential for trauma → limited to **anterior strictures found in infants before erect ambulation**; so defined, the **rarest**. The term is misused by some for any stricture without identifiable cause.

Concept — aetiology predicts biology: *iatrogenic and straddle strictures are focal ischaemic or crush scars suited to excision, whereas LS is a progressive skin disease that recurs in any skin used to repair it.*

Diagnosis and Evaluation

- **Presentation:** obstructive voiding symptoms; urinary tract infection (prostatitis, epididymitis); retention — on close inquiry, **long-tolerated obstructive symptoms** preceded retention.
- **Acute retention:** catheter attempt fails → define the obstruction by **retrograde urethrogram (RUG)**. Most are dilated acutely, often not in the patient's best interest; when in doubt, define the stricture and place a **selective suprapubic cystostomy** to buy planning time.
- **Blind passage of filiforms and blind dilation are condemned.** Flexible endoscopy is almost universally available: visualise the stricture and pass a **guidewire under direct vision**.
- **Four determinants: location, length, depth and density of spongiofibrosis.** Length and location: radiography, urethroscopy, ultrasonography. Depth and density: examination, contrast appearance, **elasticity on urethroscopy** — difficult to measure objectively.

EVALUATION MODALITIES

Modality	Contribution	Caveat
Dynamic RUG + voiding cystourethrogram (McCallum criteria)	Location and length; steep lateral oblique RUG accurately estimates length	Must be dynamic, not static ; more than one projection may be needed; done by or under the treating surgeon
Sonourethrography (urethra filled with jelly or saline)	Augments contrast studies; accurate for length of narrow-calibre annular segment	May not show the full length of spongiofibrosis
Flexible urethroscopy (local anaesthesia)	Appearance, compliance; paediatric scope inspects proximal to narrowing without dilating it	Pass to the stricture only; dilation at first evaluation usually not beneficial
Magnetic resonance imaging (MRI)	Invaluable for urethral tumours ; defines alignment of ends in pelvic fracture distraction defect	Not beneficial for routine strictures (authors)

Modality	Contribution	Caveat
Combined contrast + antegrade endoscopy	Patient with suprapubic tube who cannot void	—

- **Contrast technique:** extravasation is possible in an inflamed urethra even with gentle technique → use **contrast suitable for intravenous injection**, neat or diluted per manufacturer. **Do not thicken contrast with lubricating or anaesthetic gel** — problematic, no radiographic gain, no added comfort.
- **At surgery, evaluate the whole urethra proximal and distal to the stricture** with endoscopy and bougienage so all involved urethra is included.
- **Hydraulic trap:** voiding pressure may hold proximal segments open; once the narrow segment is relieved, unrepaired proximal disease contracts. Treat any abnormal proximal area with suspicion — **normal compliance = presumed uninvolved; coning down = involved in scar.**
- **Urethral rest:** in selected patients a suprapubic tube defunctionalises the urethra; after **6–8 weeks** the hydrodilated proximal urethra contracts, allowing precise length estimation.

Figure 86.2 — Dynamic urethrography (after McCallum, 1979)

- (A) Dynamic retrograde and (B) dynamic voiding urethrogram with McCallum's criteria, labelling bladder neck, prostatic urethra, verumontanum, membranous, bulbous and penile urethra, penoscrotal junction and fossa navicularis; (C) normal RUG; (D) normal voiding study.

Figure 86.3 — Contrast combined with endoscopy

- RUG shows a totally obliterative proximal bulbar process → voiding study suggests a wide-calibre annular area proximal to it → endoscopy through the suprapubic tube clarifies proximal anatomy and the length of obliteration.

Concept — *the imaging question is not where the narrowing is but how much spongiosum is scarred; hydraulic distension conceals proximal disease, so true extent is established by urethral rest and on-table calibration, not by the narrowest point.*

Decision Making

- First decide **palliative versus curative**. Requires: full radiological and endoscopic characterisation; knowledge of techniques and their limits; **patient and surgeon agreement on the desired outcome** and long-term expectations.
- **Curative procedures when possible and desired; palliation acceptable if it is the recognised goal.**

Treatment

- Progress over the last 50 years allows many complex strictures to be **reliably reconstructed in one stage**.
- **The reconstructive ladder** (simplest procedure first, repeated after failure, before escalating) **is archaic**.
- Discuss the goal and anticipated cure: some prefer **periodic dilation** (office, home or hospital); others want cure → surgery. Many procedures give short- and midterm success approaching long-term **>90–95%**.

Dilation

- **Oldest and simplest;** may be **curative for an epithelial stricture without spongiofibrosis**.
- **Goal:** stretch the scar without producing more scar. Bleeding means the stricture was torn, not stretched — further injury.
- **Least traumatic: soft techniques over multiple sessions.** Safest current method = **balloon dilating catheters** (filiform tip, over a guidewire, or integral Coudé tip); for initial dilation, **balloon over a wire passed under endoscopic control**.

- In selected patients, short- and midterm efficacy **equals internal urethrotomy** — but only by retrospective comparison; **no randomised trial**.
- **Optilume drug-coated balloon** — Food and Drug Administration (FDA) approval 2023 for anterior stricture; **elutes paclitaxel**, inhibiting fibroblast growth and scar formation.
- **ROBUST III** (phase III; strictures after **multiple failed endoscopic treatments**): at 1 year, stricture-free **74.6% vs 26.8%** ($P < 0.0001$); International Prostate Symptom Score (IPSS) **9 vs 19.9**. An improvement over serial endoscopy; **role as initial (rather than salvage) therapy and cost-worth undetermined**.

Internal Urethrotomy

- **Definition:** any procedure opening the stricture by **transurethral incision** — through scar to healthy tissue, releasing scar contracture so the lumen heals enlarged.

PRIMARY VERSUS SECONDARY INTENTION

Primary intention	Secondary intention (internal urethrotomy)
Epithelial edges apposed	Scarred epithelium deliberately separated
No race	Race: epithelialisation (from edges, slowing with distance) versus wound contraction (which closes the defect — not to be confused with scar contraction)
—	Success if epithelialisation completes before contraction narrows the lumen; recurrence if contraction wins

- **Predictor:** the **greater the percentage luminal narrowing, the worse the outcome**, cut-off **74% narrowing**.
- **Incision position:** the traditional **single 12 o'clock** cut is questionable — the anterior spongiosum is **thinnest from 10 to 2 o'clock**. In the bulb the urethra-corpora distance is short: a deep 12 o'clock cut enters the **triangular ligament and intracrusal space**. Distally the anterior spongiosum is thicker, but a deep cut enters the **corpora cavernosa** → **erectile dysfunction (ED)** from local cavernosal veno-occlusive dysfunction; vigorous 10 and 2 o'clock bulbar cuts carry the same risk.
- **Deep spongiofibrosis** → **cure by urethrotomy impossible**, so deep incisions are unnecessary.
- **Complications:** commonest = **recurrence**; bleeding (**almost always with erections** soon after); irrigant extravasation into perispongiosal tissue → **use normal saline**; deep incisions → **spongiosum-cavernosum fistula**, veno-occlusive dysfunction (now rare).
- **Outcome literature is poorly characterised** — stricture features under-reported, and "success" often means temporary relief. **Long-term actuarial cure is low; no success in non-bulbar strictures; a second urethrotomy adds virtually nothing**.
- **Prior dilations and urethrotomies diminish the success of later open reconstruction**. Urethrotomy success does not equal open reconstruction; **no cost analysis supports multiple endoscopic interventions**.

INTERNAL URETHROTOMY SELECTION

Favourable (good moderately long-term success)	Unfavourable
Bulbar	Penile, fossa, any non-bulbar
<1.5 cm	Longer
No dense, deep spongiofibrosis	Dense deep spongiofibrosis (e.g. straddle injury)
First procedure	Repeat urethrotomy; greater luminal narrowing

MEASURES TO OPPOSE WOUND CONTRACTION AFTER URETHROTOMY OR DILATION

Measure	Mechanism / regimen	Verdict
Prolonged Foley catheter	Mould urethra round catheter, 6 weeks	No better than 3–7 days ; 6 weeks insufficient
Intermittent self-catheterisation / home obturation	Catheter 3–5 days → obturation, frequent early, tapering over 3–6 months	Anecdotal gain; stricture recurs when obturation stops ; valid management with a dilating regimen in a motivated patient
Colchicine	Binds tubulin (? blocks wound contracture)	Non-randomised benefit only
Mitomycin C	Submucosal injection; antifibroblast, anticollagen	Decreases recurrence
Laser urethrotomy	Various lasers	Mixed; no improvement over cold knife
Removable stent	Designed to avoid epithelial incorporation; left 6 months–1 year	Success in small series (Israel)
Memokath	Removable nitinol stent	Not available in the United States; variable success
UroLume	Permanent implantable stent	Withdrawn ; many patients still have them and need treatment
Optilume	Paclitaxel-eluting balloon	ROBUST III benefit after failed endoscopy

Concept — *every endoscopic adjunct tries to tip the race between epithelialisation and wound contraction; none removes spongiofibrosis, which is why endoscopy cures only the short, superficial bulbar stricture and why repeating it simply spends the patient's reconstructive capital.*

Open Reconstruction

- Curative intent → **anastomotic versus substitution urethroplasty**. No single procedure is best for all strictures; **stratify by length, cause and site**.

Excision and reanastomosis (Russell, 1914).

- **Most dependable anterior technique: complete excision of fibrosis with primary anastomosis of normal ends.** Three technical rules: **fibrosis totally excised; anastomosis widely spatulated → large ovoid anastomosis; tension-free.**
- Success depends on **vigorous mobilisation of the corpus spongiosum**, plus dissection of Buck's fascia (compliance), **development of the intracural space**, and detachment of the bulbospongiosus from the perineal body.
- **Length: 1–2 cm easily excised; 3–5 cm in some cases.** Very proximal short bulbar strictures: **dissect the membranous urethra. The closer to the membranous urethra, the longer the excisable stricture.** Younger patients have more compliant tissue, stretching the limits.
- Many proximal strictures: **single-layer anastomosis** preferable. Length precluding total excision → **tissue transfer**.

Figures 86.4–86.6 — Excision and primary anastomosis (from Jordan, 1992)

- **86.4:** bulbospongiosus released from the perineal body, **arteries to the bulb not divided**, urethra mobilised distally; with intracural development this **shortens the urethral path by about 1–1.5 cm**; spatulated anastomosis.
- **86.5:** very proximal stricture — define → excise → **spatulate both ends dorsally** → anastomose; facilitated by membranous urethral dissection.

- **86.6:** customary division at the **membranous-bulbar juncture**, proximal vasculature ligated and divided, urethra divided at the distal limit of the membranous urethra.

Vessel-sparing technique.

STANDARD VERSUS VESSEL-SPARING ANASTOMOTIC REPAIR

	Standard (transecting)	Vessel-sparing transecting (Jordan, 2007)	Non-transecting (Andrich and Mundy, 2012)
Bulbar arteries	Ligated and divided	Preserved	Preserved
Spongiosum	Transected	Urethra divided at membranous-bulbar juncture	Not disrupted
Steps	Excise, spatulate, anastomose	Divide triangular ligament → develop intracural space → develop plane between membranous urethra and proximal vessels → excise → dorsal spatulation of membranous and proximal bulbar urethra → anastomose	Longitudinal dorsal stricturotomy → excise stricture from within → ventral urethra reapproximated → stricturotomy closed horizontally
Use	Most strictures	Proximal bulbar	Proximal strictures

- **Figure 86.7:** anastomosis with alternating polydioxanone (PDS) and poliglecaprone (Monocryl) sutures; any acceptable absorbable suture will do.
- **Advantages:** protects patients whose **distal (retrograde) supply is compromised** — trauma, previous surgery, **hypospadias**; theoretically less ED and **less cuff erosion if an artificial sphincter is later needed**. **Excellent early results; advantages not yet proven.**

Concept — *standard excision turns the bulb into a flap perfused retrogradely from the glans; sparing the bulbar arteries protects the patient whose retrograde flow cannot be trusted.*

Graft onlay.

GRAFTS FOR URETHRAL RECONSTRUCTION

Graft	Use
Oral mucosal graft (OMG) — cheek (buccal mucosal graft, BMG), lip (labial), undersurface of tongue (lingual)	Primary one-stage onlay; staged repair (LS, hypospadias); perineal urethrostomy
Full-thickness skin graft (FTSG)	Primary reconstruction; fossa navicularis (avoid in LS)
Bladder epithelial graft	Primary reconstruction
Rectal mucosal graft	Primary reconstruction
Split-thickness skin graft (STSG) , meshed	Staged reconstruction only
Posterior auricular FTSG (Wolfe graft)	Staged reconstruction (hypospadias)

- Grafts were almost abandoned for flaps; **resurgence since the late 1990s**, especially **BMG**.
- **Most successful in the bulbar urethra**, where the urethra is invested by thick, well-vascularised tissue that forms a reliable graft bed; use elsewhere, and tubed use, is increasingly reported.

GRAFT ONLAY POSITION (FIGURE 86.8)

Position	Features
Ventral	Advantageous only if spongioplasty is planned ; spongioplasty needs adjacent spongiosum relatively normal and free of fibrosis

Position	Features
Lateral	Urethrostomy through thinner lateral spongiosum → less bleeding, maximal exposure ; graft quilted to the underlying muscle bed in the bulb (authors' past preference)
Dorsal	Monseur (1980): dorsal urethrostomy, edges sutured open to triangular ligament and/or corpora. Barbagli (1995) modification: graft spread-fixed to triangular ligament and/or corpora, stricturotomy edges sutured to graft — excellent results

- **Dorsal versus ventral:** some data report superior dorsal results; others show **no difference**.
- **Figure 86.9 (Barbagli dorsal onlay):** spongiosum detached from triangular ligament and corpora → dorsal urethrotomy → graft spread-fixed to corpora with **pie-crusting** → stricturotomy edges sutured to graft and corpora.
- **Augmented anastomosis with graft (Figure 86.10):** ventral or dorsal onlay combined with excision and strip anastomosis — spongiosum detached → spongiofibrotic area marked, **narrowest segment excised**, ends spatulated dorsally → **two-layer floor strip anastomosis** → graft spread-fixed to corpora (pie-crusting, mattress sutures) → stricturotomy edges sutured to graft and corpora. **Proximal vessel-sparing variant:** excise scarred epithelium and underlying spongiosum **without dividing the spongiosum completely**.

Staged graft repair.

- **Stage 1:** medium-thickness mesh STSG, BMG or Wolfe graft placed **on dartos fascia**. Placed directly on tunica albuginea or corpora it cannot be mobilised, making tubularisation difficult — although **a midline strip adherent to the corpora is advantageous**.
- **Stage 2: tubularisation.** Schreiter and Noll (first described mesh STSG) proceeded at **3–4 months**; authors wait **12 months after STSG**.
- Useful in select cases; in the United States confined to the **most difficult cases**, single stage for most. Staged buccal and posterior auricular grafts succeed in **complicated hypospadias**; staged buccal grafts in **LS** with midterm follow-up.
- **Figure 86.11 — staged distal repair:** tight fossa stenosis extending into pendulous urethra → narrow fossa excised, stricturotomy into normal proximal urethra → buccal graft (then bolster) → **graft mature at 6 months** → Thiersch tube closed watertight, distal urethra calibrated to **about 28 Fr** → glans reconstruction and shaft closure, covered by a **tunica dartos flap carrying a parietal tunica vaginalis island** from the hemiscrotum.
- **Kulkarni single-stage panurethral repair (2012):** perineal incision, **penile invagination**, dorsal graft from proximal bulb to meatus. Good overall success, lower after a previous failed urethroplasty; **recurrences mostly proximal**.

Concept — *a graft lives on its bed, so it succeeds where the bed is thick and vascular (bulb, dorsal corpora), and is staged when the bed — or skin disease such as LS or failed hypospadias — cannot be trusted in one step.*

Flap onlay.

- Genital skin islands on the **dartos fascia of the penis** or **tunica dartos of the scrotum**. Once considered separate operations; **Quarley's microinjection studies (1983)** show them as **one concept — the skin island is a passenger on a fascial flap**, designed as in general flap surgery.
- **Three considerations:** nature of flap tissue; flap vasculature; mechanics of transfer. Skin must be **non-hirsute**; for the donor site, use **redundant non-hirsute genital skin**.

GENITAL SKIN ISLAND DESIGNS

Design	When / how	Reach
Dorsal transverse island (Duckett; Standoli 1984; Fig 86.12)	Dorsal redundancy; on dorsal dartos; lateral urethral incision	—
Ventral longitudinal island (Orandi 1972; Fig 86.13)	Ventral redundancy; vigorously mobilised on ventrolateral dartos to the perineum, or less mobilised and inverted into a pendulous defect	Pendulous or bulbar
Ventral transverse island (Fig 86.14)	Elevated on penile dartos, inverted to perineum	Proximal bulbar
"Hockey stick" (Fig 86.15)	Ventral longitudinal island extended transversely at its distal end	7–9 cm
Circular island	General penile redundancy; on entire penile dartos; most efficient ventrally based, pedicle split dorsally	Up to 15 cm
Q-flap	Circular design variant	Longer still ; complex long anterior strictures
Scrotal island	On tunica dartos, based laterally; tailor carefully (muscle component) — then no diverticulum	Reasonable in difficult cases; not first choice
Epilated midline genital island	Alternative to STSG when no non-hirsute skin; staged — epilation (needle with monopolar cautery, or epilation machine) every 6–8 weeks, repair 10–12 weeks after last	Penile dartos + scrotal tunica dartos

- **Distal strictures** (pendulous, fossa, meatus): islands advanced to the meatus by **developing glans wings or elevating the ventral glans**.
- **Figure 86.13 — penile longitudinal island**: full-thickness incision through dartos and superficial Buck's fascia lateral to spongiosum → elevate dartos flap past midline → **lateral urethrostomy** facing the flap opens the stricture → develop skin paddle → fix medial edge to stricturotomy → **invert flap** into defect → watertight subepithelial running absorbable monofilament line → skin closed.
- **Figure 86.15 — pedicle**: Buck's and dartos fascia (denuding tunica of spongiosum and corpora), based on **superficial external pudendal and internal pudendal vessels** in the scrotum; can reach **any area of the urethra**, passed through a subscrotal tunnel to a laterally placed urethrostomy.

FOSSA NAVICULARIS RECONSTRUCTION

- Repeated dilation or sequential meatotomy seldom durable → random penile skin flaps into the meatotomy (function better, **cosmesis poor**) → **dartos-based skin islands: excellent function and cosmesis**. Design accounts for **penile hair** and transfer mechanics (**transposition versus advancement**).
- **Jordan (Fig 86.17)**: ventral urethrotomy through stenosis → **transverse ventral island** on ventral dartos → **transposed and inverted** into meatotomy defect. **De Sy (Fig 86.18)**: ventral longitudinal island **advanced** into the defect, part of the flap de-epithelialised.
- FTSG has been used for the fossa; **skin grafts inappropriate in LS where avoidable**, and skin islands in LS are questioned.

ONLAY VERSUS TUBULARISED

- **Onlay (graft or flap) outperforms tubularised grafts or skin islands — avoid tubes**. If unavoidable, **limit tube length** by aggressive mobilisation and excision. **Tubularised flaps outperform tubularised grafts**.

- Extremely long strictures: **flap distally + graft onlay proximally**. Where a tube is required, **graft spread-fixed to re-establish a urethral plate + flap onlay** appears better than a tubed flap, even onlay-tube-onlay (small series, short follow-up).

Concept — *Quartey unifies all flaps: the skin is a passenger and the dartos pedicle is the operation; the island is chosen by where the hairless redundant skin lies and how far it must travel.*

Augmented anastomosis.

- Combine **excision of the stricture with skin island onlay (Fig 86.16) or graft onlay (Fig 86.10)**. Very narrow (near- or totally obliterative) segments are excised completely → **roof or floor strip anastomosis** → remaining urethrotomy filled with graft or island.
- **Figure 86.16 — circular island:** island elevated with patient flat → onlay begun, flap passed into perineal dissection, penis closed → repositioned to lithotomy → flap retrieved, narrow segment excised, dorsal spatulation → onlay completed, floor strip anastomosis closed.

DURABILITY AND TRADE-OFFS OF OPEN TECHNIQUES

Technique	Durability	Notes
Excision and primary anastomosis	Durable ; success at 1 year does not decay at the same rate	Authors' long-term series: recurrence rare, mostly early (first 6 months); de novo ED low , higher after severe straddle injury
Tissue transfer (graft or flap)	Excellent at 1 year, deteriorates with longer follow-up	—
Graft onlay vs flap onlay	Equivalent (meta-analysis)	Grafts technically far easier ; flaps may be superior in radiation strictures, multiply operated patients, pendulous strictures
Tubularised repair	Worse than onlay	Flap tube better than graft tube

- **Postoperative ED:** overall rate after urethroplasty ≈ **that after circumcision**; **longer reconstructions carry higher risk**; function often improves with time.
- **Lichen sclerosus:** flaps were embraced → **very high recurrence**; staged **skin** grafts also recurred. Because LS is a **skin disease**, skin as flap, single-stage or staged graft can itself become involved → **staged oral mucosal grafts** advocated; short-term results better, long-term unavailable. Fossa stricture with LS repaired with ventral transverse island: **high recurrence**.

Concept — *excision and primary anastomosis is the gold standard because a healed anastomosis does not decay, whereas transferred tissue — above all skin in LS — carries its own continuing failure mode.*

Perineal urethrostomy (PU).

- **Complex strictures** — LS, failed reconstructions, failed hypospadias — have long histories of instrumentation and surgery, **diminished sexual function and quality of life**, and arrive late at referral centres.
- **Pananterior multigraft repair** (one or more stages) is feasible with fair success but may not be optimal; **elderly high-risk patients** may not warrant it; even in the young, staged success falls after **multiple failed urethroplasties** → **PU: simple and effective**.
- **Challenges:** disease into the **proximal bulbar or membranous urethra** defeats traditional PU; **prior pelvic radiation and LS** increase PU restructure.
- **BMG PU:** brings the urethrostomy to the perineal surface instead of burying skin to the urethral opening; a graft in the neomeatus **reduces circumferential scarring**.

BMG PERINEAL URETHROSTOMY — SEQUENCE

1. High lithotomy → **inverted U incision** → bulbospongiosus divided in the midline.
 2. Spongiosum freed from triangular ligament proximally and corpora distally → **transected transversely** → **dorsal urethrotomy** until healthy mucosa.
 3. Healthy **ventral urethra brought to perineal skin** (not a channel of inwardly mobilised skin) → proximal calibration to **30 Fr bougie à boule** → distal urethra closed, allowed to retract.
 4. **BMG** long enough to reach from dorsal urethrotomy to skin completes the dorsal wall; graft bed = corporal bodies; **fix graft to corpora** so neomeatal skin cannot pull it off its blood supply.
- **Outcome:** panurethral strictures of mixed cause (unknown commonest; LS, iatrogenic, failed hypospadias, Fournier gangrene, urethral and penile cancer) — **most successful at midterm**; recurrences managed by revision or periodic dilation. **Reproducible, viable alternative** in selected patients.

Concept — PU trades anatomy for reliability: when the anterior urethra has exhausted its reconstructive options, the diseased segment is bypassed; the BMG corrects the one weakness of a deep PU — circumferential skin-to-urethra scarring.

KEY POINTS: TREATMENT OF URETHRAL STRICTURE

- Agree the **goal of treatment** before choosing it.
- **Dilation** — oldest, simplest; goal is atraumatic stretching; **seldom curative** (only the epithelial stricture without spongiofibrosis).
- **Internal urethrotomy** — reserve for **bulbar, <1.5 cm, no dense deep spongiofibrosis**; repeated dilation or urethrotomy diminishes later open success.
- **Lasers** — no advantage over cold knife.
- **Excision and primary anastomosis is the gold standard**; once thought limited to <1.5–2 cm, longer strictures are now excised with better anatomical understanding.
- **Grafts = flaps** in results (meta-analysis); flaps more complex; **augmented anastomosis** often better than pure onlay; all flaps are one operation with multidimensional application.
- **PU** is viable in selected patients; **BMG** helps very proximal bulbar disease.

KEY POINTS: URETHRAL STRICTURE

- "Stricture" = **anterior** disease of epithelium and spongiosum; posterior lesions = **pelvic fracture urethral injuries (PFUIs)**; prostatic urethra and bladder neck = **contractures or stenoses**.
- Bulbar urethra eccentric (dorsal) in spongiosum; genital skin **dual, bilateral, fasciocutaneous** supply; spongiosum from the **common penile artery**.
- **Idiopathic and iatrogenic** causes predominate, then trauma; gonococcal strictures now rare; **LS strictures behave as inflammatory**.
- Obstructive symptoms or infection; retention follows long-tolerated symptoms.
- Define **length, location, depth, density** — contrast, endoscopy, selective ultrasonography; examine the whole urethra; **paediatric cystoscope** useful.

PELVIC FRACTURE URETHRAL INJURIES

- Result of **blunt pelvic trauma**; accompany **about 10% of pelvic fractures**.

- **Straddle injury** can totally disrupt the urethra but usually involves **only the bulbar urethra** (spongiofibrosis may still obliterate it). **Distraction injuries are unique to the membranous urethra.**
- **"Apple off its stem" analogy** implies injury at the prostatic apex — **wrong: the commonest point is the departure of the bulbar from the membranous urethra**; any part between that point and the apex may be involved.

AGE AND SITE OF DISTRACTION

Postpubertal	Prepubertal
Prostatic urethra seldom involved	Fragile prostatic urethra — injury can extend into it

- **Many injuries are incomplete** — a **strip of epithelium** remains. An **aligning catheter** (easily placed with flexible endoscopy) may let the urethra heal unscarred or with an easily managed stenosis; if complete, it aligns the ends and **facilitates later reconstruction**. Some centres evaluate acutely by endoscopy alone, expediting the whole process.
- An **aligning catheter is a guide, not traction** on bladder and prostate; it may also **drain the liquefying pelvic haematoma** and hasten its resolution.
- **Close follow-up after the voiding trial** — many develop stricture after catheter removal and need definitive repair. Primary realignment has been linked to **delayed repair and more failed endoscopic treatments** → some advise against routine use.

Evaluation

- Define **depth, density, length and location**. In PFUI **depth and density are predictable**. Location affects continence → a **counselling factor, not a determinant of approach**. **Length must be determined as precisely as possible.**
- **Contrast studies are first line**. Cystogram: bladder outline and **rostral displacement of the proximal urethra**; absent contrast in the posterior urethra says little (inconclusive) about bladder neck integrity.
- If the patient can relax to void and the cystogram fills the posterior urethra, a **simultaneous RUG** shows the defect length — the **exception, not the rule**. The RUG's main role is to show whether the **anterior urethra is normal** — if so, **successful anastomotic repair is ensured**.
- Primary anastomosis is possible **even with some anterior involvement** and after a **previous failed posterior repair** (slightly higher failure). **Primary anastomosis is the goal in all until proven impossible.**
- Proximal urethra not seen → **antegrade flexible endoscopy through the suprapubic tract + RUG**: assess the bladder neck, advance to the obstruction; the unopacified gap is the defect.
- **Bladder neck appearance** on contrast or antegrade endoscopy **does not predict its eventual function**.
- **MRI**: authors' experience minimal; studies show benefit in difficult cases — when the **prostatic apex and urethral stump lie more cephalad, infrapubectomy** is more often needed.

Repair

- **Timing**: set by associated injuries — ideally **3–6 months after trauma**; lower-limb orthopaedic injuries often delay it.
- **Most PFUIs are short** → mobilisation of spongiosum + **primary anastomosis**: spatulated proximal anterior urethra to **widely spatulated membranous urethra** for epithelial apposition.
- **About 10% are complex**, may have fistulae (**commonest urethrorectal**) — technically more demanding.

APPROACHES TO POSTERIOR URETHRAL RECONSTRUCTION

Approach	Role	Drawbacks
Perineal (anastomotic)	Almost all PFUIs , including most difficult; adults and most children	Exposure harder in children
Transpubic / abdominoperineal (Waterhouse 1973)	Unnecessary for distraction injuries (authors); some still rely on it	Pubectomy: penile shortening, erection destabilisation, pelvic instability with exercise-related chronic pain
Above-and-below	When concomitant bladder neck surgery is planned	—
Posterior sagittal trans-sphincteric	Proposed for children — elegant exposure	Tempts substitution repair where anastomosis is possible and superior
Endoscopic	Not for curative repair	—

- **Bladder neck:** competence cannot be judged before continuity is restored — contrast may fill the prostatic urethra with a competent neck; an obviously scarred neck at endoscopy is often continent afterwards; **scar-incarceration incontinence is infrequent**. Practice: **restore continuity, forewarn if concerned, treat incontinence later**; simultaneous bladder neck and posterior urethral reconstruction is not usually undertaken.

PERINEAL ANASTOMOTIC REPAIR — SEQUENCE

1. **On-table flexible endoscopy** via meatus and suprapubic sinus — excludes **bladder stones**.
2. Haygrove or van Buren sound (or cystoscope; hollow Haygrove — Gelman sound — admits a flexible scope to view the bladder neck) → **feel the proximal stump impulse**.
3. **Exaggerated lithotomy** (Fig 86.19): Allen or Guardian stirrups; no pressure on lateral legs or calves; hips raised by elevating the buttock section; boots set to avoid **common peroneal nerve stretch**; minimal hip and knee flexion.
4. Incision **anterior to the transverse perineal muscles** (anterior perineal triangle — versus posterior anal triangle for perineal prostatectomy); **λ-shaped incision** in larger patients → midline muscle fusion over the bulb → under the scrotum → ring retractor.
5. Midline muscle fusion divided, muscle dissected off the bulb → spongiosum detached from triangular ligament and corpora; bulb detached from perineal body → dissection to infrapubic space and through fibrosis.
6. **Bulbar arteries:** sharp-tipped haemostat and monopolar cautery; **avoid suture ligature** (nerves nearby entering the corpora); a **vessel-sparing PFUI repair** is sometimes possible.
7. **Divide the triangular ligament; develop the intracural space to the pubis;** ligate the dorsal vein; beware **cavernosal or dorsal arteries displaced into the space** by trauma (Doppler if in doubt); periosteal elevator on the retropubic surface frees tissues.
8. **Haygrove staff** via suprapubic sinus through the bladder neck → palpate → **resect fibrosis to normal planes** (histology) → open epithelium over the staff tip → endoscopy confirms the distal limit of the posterior urethra.
9. If tension: **mobilise spongiosum beneath the scrotum as the last manoeuvre** (retrograde supply may be tenuous); detaching Buck's fascia increases compliance and limits the need.
10. Spatulate proximal urethra to accept **≥30 Fr bougie à boule** → **8–10 tagged 4-0 PDS sutures** (Heaney and Ravitch drivers help) → spatulate distal to **30 Fr** → **soft silicone ribbed stent** under vision → irrigate clot → seat anastomosis.

11. Reattach spongiosum to corpora and bulb to perineal body → **two suction drains** (deep to the muscle and Colles fascia closure; superficial beneath the subcutaneous closure).
- **Chordee**: avoid by **not mobilising beyond the penoscrotal junction**; counsel that mild chordee is possible — rare, usually not sexually disabling, a **fair trade** for a primary anastomosis.
 - **Progressive length manoeuvres** shorten the path so chordee is avoided: **intracrusal space development** → **infrapubectomy** → **rerouting of the spongiosum**.
 - **Rostral distraction**: be prepared for **infrapubectomy** (Fig 86.22 — inferior symphysis resected with a **Kerrison rongeur**, as much as needed), **corporal rerouting** (Fig 86.23), or both; palpate the bladder neck and sound before sub-pubic dissection; if tension persists over the penile hilum, extend the infrapubectomy under one corpus and **reroute the urethra lateral to the crus**.
 - **Figure 86.20 (perineal repair)**: Colles fascia opened → muscle-bulb plane developed → bulb exposed → retractor exposes inferior fascia of the urogenital diaphragm → fibrosed urethra incised, bulb freed → anterior urethra opened → Haygrove staff passes into the perineum once the fibrotic defect is resected. **Figure 86.21**: incising the triangular ligament shortens the arc; symphyseal periosteum mobilised for retractors; crura displaced to expose and ligate the dorsal vein; wide exposure of periapical fibrosis.

Postoperative Management

- Small soft silicone stent; **urine diverted suprapubically, urethral catheter plugged — a stent only**. Overnight stay; discharge on an **anticholinergic**; **suppressive antibiotic only if the preoperative culture was positive**.
- **Contrast voiding trial at 21–28 days**: stop oxybutynin **24 hours** before; straightforward anastomosis at **21 days**, rostral distraction **3–5 days later**. Catheter out → bladder filled with contrast → void; check **no extravasation, widely patent. No pericatheter RUG**.
- **Flexible endoscopy at about 6 months** — reconstruction mature and widely patent; endoscopy has almost replaced retrograde studies.
- **Curative rates in the high 90s %**. In large centres, **failures are not technical anastomotic restenosis but ischaemia of the proximal spongiosum** → stenosis of the mobilised segment, which has become a **flap on retrograde dorsal arterial flow through the glans**.
- **At-risk groups** studied by pudendal angiography: dorsal penile nerve injury, failed repair elsewhere, **lateral-impact** and **"windswept"** pelvic fractures. Many had pudendal lesions; most had **vascular reconstitution**.

INTERNAL PUDENDAL ARTERIAL STATUS AFTER PFUI

Arterial status	Potency	Reconstruction	Duplex	Action
Intact on ≥1 side	Often potent	Reliably cured	Normal	Reconstruct
Reconstituted only (unilateral or bilateral)	Never potent	Reliably reconstructed	Never normal	Reconstruct; candidate for penile revascularisation for potency
Bilateral complete occlusion	—	Only group at risk of ischaemic stenosis	Abnormal	Revascularise first , then reconstruct

- **Algorithm: duplex ultrasonography first** → **angiography only if duplex abnormal**; normal duplex predicts a good result.
- **ED after PFUI** is mostly a **consequence of the injury**, sometimes of surgery. Pudendal arterial injury is **under-recognised**, and much ED is **vascular**; a portion is **neurogenic** (normal arterial inflow).

Figure 86.24 — Deep penile vasculature

- **(A) Normal:** common penile artery flow to the glans gives **retrograde** spongiosal flow; intact bulbar arteries add **antegrade** flow.
- **(B)** Bulbar arteries divided + mobilisation → **all spongiosal flow retrograde**.
- **(C) Hypospadias:** distal spongiosum interrupted; after bulbar division, retrograde supply may be inadequate even with intact common penile flow → **ischaemic stenosis**.
- **(D) Common penile artery injury + bulbar division** → **ischaemic necrosis or stenosis**.

Summary

- **Almost all PFUIs can be reconstructed perineally by anastomosis; endoscopic techniques should not be used for curative repair.**
- **Children:** same goals; most reconstructed through the same perineal exposure (harder).
- **Vessel-sparing** repair extended to select PFUIs — feasible, good results in small numbers, **advantage unproven**.

Concept — *PFUI repair is technically an anastomosis but biologically a flap: failure comes from bulbar ischaemia, not the suture line, so pudendal inflow is screened before — not after — the spongiosum is mobilised.*

KEY POINTS: DISTRACTION INJURIES OF THE URETHRA

- PFUIs follow blunt pelvic trauma, in about 10% of pelvic fractures; often **incomplete**, with an epithelial strip left.
- **Acute aligning catheter is controversial**, but at worst it facilitates later reconstruction.
- Define anatomy with **contrast + endoscopy ± selective MRI**; bladder neck appearance does **not** predict function; simultaneous bladder neck and posterior urethral reconstruction is not usually undertaken.

VESICOURETHRAL ANASTOMOTIC STENOSIS

- **Robotic-assisted radical prostatectomy (RARP)** has **reduced vesicourethral anastomotic stenosis (VUAS)** — fewer leaks, better mucosal apposition, running anastomosis under magnification. Some still develop an **obliterative distraction defect** or severe stenosis.
- **Length:** simultaneous cystogram + RUG, and/or RUG + **antegrade endoscopy** through the suprapubic tube.
- Patients often comorbid, with **thick small bladders** and adverse **body habitus**, both contributing to the original difficulty. **An indwelling suprapubic tube is always an option**, including long term.
- **Endoscopic (short distractions):** aggressive incisions at **3 and 9 o'clock**, repeated at about **3 weeks**; holmium laser versus cold knife debatable; **hot knife unnecessary**. **If continuity must be "cored through", endoscopy is inappropriate**. Radial urethrotomy + **intralesional mitomycin C**: good initial success in recalcitrant stenosis.
- **Continent catheterisable augmentation** may be better than aggressive functional reconstruction (channel difficult in obesity); **diversion is a primary option** when functional reconstruction is not obvious.
- **Omentum interposition** has been used for distraction defects and complex fistulae, allowing safe mobilisation of the rectum from the scar or fistula.

FUNCTIONAL RECONSTRUCTION OF VUAS

	Robotic	Perineal
Level	Above the external urethral sphincter	Through membranous urethra / external sphincter
Technique	Redo anastomosis (favoured) → bladder flap → BMG (last option)	High lithotomy; dissection as for PFUI; vessel-sparing attempted (artificial urinary sphincter, AUS, likely later); partial infrapubectomy occasionally
Continence	Some adequate for lifestyle; others AUS	High incidence of incontinence — counsel
Evidence	Promising; small numbers, minimal follow-up	—

Concept — *the level of the stenosis relative to the external sphincter decides both the approach and the continence cost; beyond that the choice is between restoring the channel and abandoning it for a tube, augmentation or diversion.*

COMPLEX FISTULA OF THE POSTERIOR URETHRA

- **Rectourethral and prostatorectal fistulae are increasing with multimodal prostate cancer treatment;** rare after prostatectomy alone.

SIMPLE VERSUS COMPLEX FISTULA

	Simple	Complex
Definition	<2 cm, no radiation, no infection	>2 cm, or radiation, infection, or immunocompromise
Faecal diversion	At the time of repair (diversion first, then repair)	Before definitive repair
Approach	Perineal; or York-Mason (posterior sagittal) — excellent for simple non-irradiated fistula	Anterior transperineal with muscle interposition ± BMG

- **Plan around future function:** urinary function doubtful → **urinary diversion**; bowel function doubtful → **more permanent bowel diversion**; both adequate → **repair**.
- **Perineal approach is commonest** (urologists' familiarity); York-Mason is unfamiliar to urologists.
- **Principles of both:** excise the fistula → remove any foreign body → **tension-free closure** → **vascularised flap** if needed.
- **Transperineal series with gracilis interposition ± buccal graft:** non-irradiated fistulae — **all closed**, no later strictures, most bowel undiverted; irradiated — most closed in one stage and nearly all ultimately, but **more recurrent strictures** and substantial **permanent bowel diversion**. **Muscle interposition integral; BMG invaluable** for urethral closure.
- The chosen approach **facilitates or limits** options for bowel, urethra and tissue interposition.

Concept — **radiation is the dividing line:** *without it a fistula is a closure problem solved by excision and interposition; with it, it becomes a function problem in which diversion is often the honest endpoint.*

KEY POINTS: VESICourethRAL DISTRACTION DEFECTS AND COMPLEX FISTULAE

- Vesicourethral distraction defects are a **complication of radical prostatectomy**.
- **Suprapubic tube** is always a long-term option; **continent catheterisable augmentation** may beat aggressive reconstruction.

- **Omentum interposition** allows safe rectal mobilisation from scar or fistula.
- **After radiation:** let radiation effects settle; **tissue interposition is the rule**; functional reconstruction often impossible; some regard **diversion as safest**.
- Plan around **ultimate urinary and bowel function**.